Sina Abdipoor (思纳), Ph.D.

AI Engineer, Researcher, and Teacher specializing in Operations Research, Optimization, and Machine Learning

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Summary

I am a Lecturer in the Department of Data Science and Artificial Intelligence at Sunway University, Malaysia, where I lead ScholORs, an optimization research group. I hold a Ph.D. in Computer Science from Universiti Putra Malaysia.

My research is in optimization and Operations Research, with a focus on metaheuristics and hybrid algorithms for combinatorial and stochastic problems such as university timetabling. A central question in my work is generality, that is, how well an algorithm performs across problems rather than on a single benchmark. My interests also include game theory and multi-agent systems. My work appears in journals such as IEEE Access, Intelligent Systems with Applications, and Applied Computing and Informatics, and I regularly take part in GECCO and PATAT. Through ScholORs, I sometimes work with other experts in the field, such as the optimization research group at Lancaster University. I also regularly collaborate with experts from all over the world, such as researchers at Hefei University in China. I review for Expert Systems with Applications, Applied Soft Computing, and IEEE Transactions on Intelligent Transportation Systems.

I have taught since 2016. At Sunway, I teach optimization, applied machine learning, and software engineering, serve on the program committee of the Master in Artificial Intelligence, and help design its curriculum and new courses. I supervise Master's students and more than 20 undergraduate research projects.

From 2020 to 2025, I also worked as an independent Operations Research consultant, building optimization and prediction models for organizations. This experience keeps my research close to real problems.

Research Interests

Research: Optimization, Operations Research, Artificial Intelligence, Game Theory, Multi-Agent Systems

Teaching: Optimization, Search Algorithms, Evolutionary Computing, Applied Machine Learning, Software Engineering, Programming, Game Theory, Multi-Agent Systems, Artificial Intelligence in Sustainability

SDGs: SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production)

Education

Ph.D. **Universiti Putra Malaysia** 2020 – 2026
Computer Science
Thesis: Optimality and Generality of Metaheuristics in University Course Timetabling
- Proposed a hybrid metaheuristic for University Course Timetabling with Student Sectioning (UCTSSP), tested on the International Timetabling Competition 2019 benchmark.
- Supervisor: Associate Professor Dr. Razali Yaakob. Committee: Professor Dr. Salwani Abdullah, Dr. Khairul Azhar Kasmiran, Dr. Hazlina Hamdan
- Collaborators: Associate Professor Dr. Say Leng Goh, Professor Graham Kendall

M.Sc. **K. N. Toosi University of Technology** 2016 – 2019
Computer Engineering, Artificial Intelligence and Robotics
Thesis: Analyzing Parameter Tuning of Evolutionary Algorithms in Optimization Problems
- Supervisor: Associate Professor Dr. Amin Nikanjam
- Courses: Machine Learning, Reinforcement Learning, Evolutionary Computing, Pattern Recognition, Probabilistic Graphical Models, Artificial Neural Networks, Big Data

B.Sc. **Imam Khomeini International University** 2012 – 2016
Computer Engineering, Software Engineering
- Final year project: Designing and Launching a Restaurant Management System for a Local Cafe
- Supervisor: Associate Professor Dr. Morteza Mohammadi Zanjireh
- Courses: Artificial Intelligence, Data Mining, Algorithm Design, Database Design, Object-Oriented Programming, Data Structures

Academic Positions

Sunway University, Lecturer, Department of Data Science and Artificial Intelligence — Dec 2025 – present

- Head of ScholORs, an optimization research group that sometimes works with other experts in the field, such as the optimization research group at Lancaster University.
- Researching metaheuristics and hybrid optimization for real-world timetabling, with 9 papers in 2026.
- Collaborating regularly with experts from all over the world, such as researchers at Hefei University, China, with joint papers at GECCO 2026 and PATAT 2026.
- Member of the program committee of the Master in Artificial Intelligence, contributing to curriculum development and the design of new courses.
- Subject Lead for Algorithms for Optimization, and teaching postgraduate and undergraduate courses in machine learning, software engineering, and data science capstone projects.
- Supervising 2 Master's students and more than 20 undergraduate final year projects.

Universiti Putra Malaysia, Graduate Researcher & Tutor — Sep 2020 – Sep 2025

- Researched hybrid metaheuristics for university course timetabling for my Ph.D., published in IEEE Access and Intelligent Systems with Applications.
- Taught Computer Programming 2 (SSK3101) and Machine Learning (SSK5603) at undergraduate and postgraduate levels.

Publications

An integration of discrete wavelet transform and time-series recurrent neural network to improve stock price prediction — Aug 2026
Keong Kuon Lock, Razali Yaakob, **Sina Abdipoor**, Mingxuan Li
Applied Computing and Informatics
10.1108/ACI-08-2025-0335

Integrative Hybrid Local Search with Luby's Restart Strategy for University Course Timetabling with Student Sectioning — Aug 2026
Mingxuan Li, Thomas Weise, **Sina Abdipoor**, Jourdan D'orville, Say Leng Goh, Razali Yaakob, Salwani Abdullah
15th Conference on the Practice and Theory of Automated Timetabling (PATAT)
patatconference.org/patat2026/submissions/PATAT_2026_paper_1306.pdf

SpOC 4 Challenges 1 Beginner: Permutation Encodings and Challenge 3: Simulated Annealing for the Luna Tomato Advertising — Jul 2026
Mingxuan Li, Thomas Weise, **Sina Abdipoor**, Zhi-Ze Wu
Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO)
10.1145/3795101.3815558

Solving the SpOC 4 Challenge 2 with a Multi-Step Permutation Encoding for the Keplerian Traveling Salesperson — Jul 2026
Thomas Weise, **Sina Abdipoor**, Mingxuan Li, Zhi-Ze Wu
Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO)
10.1145/3795101.3815557

A Machine Learning Approach to Forecast Car Valuation in the Automotive Industry — Jun 2026
Chun-Yun Chiang, **Sina Abdipoor**, Anwar P. P. Abdul Majeed, D. O. Alebiosu, Wan Siti Nur Aiza, Muhammed Basheer Jasser
2026 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)
10.1109/I2CACIS69435.2026.11600322

A Review on Artificial Intelligence Techniques Employed for Enhancing NPC Behavior in Video Games — Jun 2026
Ruben Raj Karthigasu, Muhammed Basheer Jasser, Samuel Mofoluwa Ajibade, **Sina Abdipoor**, Na-Chua Hui
2026 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)
10.1109/I2CACIS69435.2026.11600010

Real-World University Timetabling: A Two-Stage Collaborative Hybrid Metaheuristic — May 2026
Chee Seng Leong, Say Leng Goh, **Sina Abdipoor**, Muhammed Basheer Jasser, Na-Chua Hui, Hao Luo
2026 IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
10.1109/CSPA68262.2026.11517821

A Comparison Study of Swarm Intelligence Algorithms Employed for Optimizing ANN Training in Flood Prediction May 2026
Bibi Aamirah Shafaa Emambocus, Muhammed Basheer Jasser, Samuel Mofoluwa Ajibade, **Sina Abdipoor**, Na-Chua Hui
2026 IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
10.1109/CSPA68262.2026.11517716

Urban House Price Forecasting in Malaysia Using Tree-Based Machine Learning Models Apr 2026
Ying Zheng, Hui Shen Choo, Yoke Leng Yong, Umi Najiah Ahmad Razimi, Muhammed Basheer Jasser, **Sina Abdipoor**
2026 IEEE 2nd International Conference on Robotics and Technologies for Industrial Automation (ROBOTHIA)
10.1109/Robothia68364.2026.11507118

Distance to Feasibility: Performance Assessment of Baseline Metaheuristics on University Course Timetabling with Student Sectioning Dec 2025
Sina Abdipoor, Mingxuan Li, Say Leng Goh, Razali Yaakob, Hao Luo, Salwani Abdullah
2025 13th IEEE Conference on Systems, Process & Control (ICSPC)
10.1109/ICSPC68261.2025.11325835

Optimality Versus Generality: Performance Assessment of Meta-Heuristics in Educational Timetabling 2025
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Hazlina Hamdan, Khairul Azhar Kasmiran
IEEE Access
10.1109/ACCESS.2025.3565522

Digital Twin System for Smart Campuses Oct 2024
Hui-Jie Wang, Yuanyuan Cai, Fabu Yue, Hao Luo, **Sina Abdipoor**, Say Leng Goh
2024 IEEE 10th International Conference on Control Science and Systems Engineering (ICCSSE)
10.1109/ICCSSE63803.2024.10823754

Hybrid Graph Neural Networks with LSTM Attention Mechanism for Recommendation Systems in MOOCs Mar 2024
Best Paper Award
Hao Luo, **Sina Abdipoor**, Qing Li, Li-Tao Xu
2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
10.1109/CSPA60979.2024.10525319

Smart-Kube: Energy-Aware and Fair Kubernetes Job Scheduler Using Deep Reinforcement Learning Sep 2023
Saeid Ghafouri, **Sina Abdipoor**, Joseph Doyle
2023 IEEE 8th International Conference on Smart Cloud (SmartCloud)
10.1109/SmartCloud58862.2023.00035

Meta-heuristic approaches for the University Course Timetabling Problem Sep 2023
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah
Intelligent Systems with Applications
10.1016/j.iswa.2023.200253

International Timetabling Competition 2019: A Systematic Literature Review Jun 2023
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Khairul Azhar Kasmiran, Hazlina Hamdan
2023 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)
10.1109/I2CACIS57635.2023.10193383

Awards

Best Paper Award Aug 2026
19th International Conference on Advanced Computer Theory and Engineering (ICACTE)

National Finalist, 3 Minute Thesis (3MT) Competition Jun 2024
Universiti Kebangsaan Malaysia

Winner, 3 Minute Thesis (3MT) Competition Mar 2024
Universiti Putra Malaysia

Best Paper Award — Mar 2024
20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)

Special Jury Award, Postgraduate Colloquium — Oct 2023
Universiti Putra Malaysia

Special Graduate Research Allowance (SGRA) — Jun 2021
Universiti Putra Malaysia, grant FRGS/1/2020/ICT02/UPM/02/5

Rank 18, National University Entrance Exam (Konkoor), Private Universities, Computer Engineering — May 2016
National Organization for Educational Testing (Sanjesh)

Rank 179, National University Entrance Exam (Konkoor), Governmental Universities, Computer Engineering — May 2016
National Organization for Educational Testing (Sanjesh)

Silver Medal, National Universities Swimming Championship — 2013
Imam Khomeini International University

Talks

Applied Operations Research in Driving Sustainability — China, Aug 2026
Guest talk, Hefei University

AI in Higher Education and AI Governance — May 2026
Invited interview, University of Lincoln

Preparing New UPM Ph.D. Students for 3MT 2025 — Malaysia, 2025
Sharing session, Universiti Putra Malaysia

Conference Presentations

19th International Conference on Advanced Computer Theory and Engineering (ICACTE) — China, 2026

15th Conference on the Practice and Theory of Automated Timetabling (PATAT) — United Kingdom, 2026

67th Annual Conference of the Canadian Operational Research Society (CORS) — Canada, 2026

Genetic and Evolutionary Computation Conference (GECCO) — Costa Rica, 2026

IEEE International Conference on Automatic Control & Intelligent Systems (I2CACIS) — Malaysia, 2026

22nd IEEE International Colloquium on Signal Processing & Its Applications (CSPA) — Malaysia, 2026

2nd IEEE International Conference on Robotics and Technologies for Industrial Automation (ROBOTHIA) — Malaysia, 2026

13th IEEE Conference on Systems, Process & Control (ICSPC) — Malaysia, 2025

66th Annual Conference of the Canadian Operational Research Society (CORS) — Canada, 2025

10th IEEE International Conference on Control Science and Systems Engineering (ICCSSE) — China, 2024

65th Annual Conference of the Canadian Operational Research Society (CORS) — Canada, 2024

20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA) Malaysia 2024

8th IEEE International Conference on Smart Cloud (SmartCloud) Japan 2023

IEEE International Conference on Automatic Control & Intelligent Systems (I2CACIS) Malaysia 2023

Teaching

Algorithms for Optimization (BAI3184) Sep 2026
Subject Lead, Bachelor in Artificial Intelligence, Sunway University

Capstone Project 1 and 2 (MRP5015, MRP5025) Mar 2026
Lecturer, Master in Data Science (ODL), Sunway University

Applied Machine Learning (MAI5043) Feb 2026
Lecturer, Master in Artificial Intelligence, Sunway University

Software Engineering (SEG2202) Feb 2026
Lecturer, B.Sc. Computer Science, Sunway University

Machine Learning (SSK5603) 2020
Tutor, Universiti Putra Malaysia

Computer Programming 2 (SSK3101) 2020
Tutor, Universiti Putra Malaysia

Evolutionary Computing 2018
Teaching Assistant, K. N. Toosi University of Technology

Algorithm Design 2014
Teaching Assistant, Imam Khomeini International University

Supervision

Whong Ye Wen 2026 – present
Master in Data Science, Sunway University, Main Supervisor

Tan Yi Wei 2026 – present
Master in Artificial Intelligence, Sunway University, Main Supervisor

More than 20 undergraduate final year projects 2026 – present
Sunway University

Mingxuan Li 2025 – present
B.Sc., Queen Mary University of London, Main Supervisor

Peer Review

Expert Systems with Applications, Elsevier 2025 – present

IEEE Transactions on Intelligent Transportation Systems 2026 – present

Applied Soft Computing, Elsevier 2026 – present

Service and Leadership

Sunway University, Student Engagement & Experience Committee Member, Mobility Programme in AI and Automation Sep 2026

- Hosted 15 students from Galgotias University (India), Óbuda University (Hungary), and Sunway University for a one-week programme of CDIO, AI, and Automation workshops.
- Received a Certificate of Appreciation from the Faculty of Engineering and Technology.

Putra Graduate Association, Universiti Putra Malaysia, Chief Exco of International Relations Affairs & Alumni — Jul 2022 – Jul 2023

- Organized programs for international postgraduate students and alumni.
- Coordinated postgraduate clubs and student country representatives, in Malaysia and abroad.
- Maintained the UPM postgraduate alumni records.
- Represented the association at iPutra and Education Malaysia Global Services (EMGS).

Professional Memberships

Canadian Operational Research Society (CORS) — Apr 2024 – present
Member

Professional Experience

Independent Consultant, Operations Research Engineer — Jan 2020 – Mar 2025

- Partnered with organizations to analyze workflows and find inefficiencies in resource allocation.
- Designed, proposed, and deployed optimization models.
- Presented findings and recommendations to stakeholders and management.

Pars Six Sigma Excellence, Software Engineer — Jul 2017 – Jul 2019

- Designed, built, and maintained a CRM system for employees and customers in a small team of software engineers.
- Developed the Java backend using a variety of libraries, frameworks, and APIs.
- Designed and implemented the SQL database.

Kish Institute of Science and Technology, Senior English Teacher & Supervisor — Nov 2016 – May 2019

- Taught Upper-Intermediate and Advanced English (B2 to C2) to teenagers and adults, following Oxford's syllabus.
- Led IELTS workshops, level assessments, and teacher interviews.
- Observed classes to keep teaching standards high.

IKCO, Android Developer (Internship) — Jun 2016 – Sep 2016

- Built an Android app with SQLite that lets assembly line workers check parts stock in real time and order parts without leaving their station.

ParsOnline, Network Engineer — Jun 2014 – Mar 2015

- Installed and configured network equipment.

Projects

ESA Space Optimisation Competition (SpOC 4) — 2026
Ranked 4th globally in one track of the European Space Agency's space logistics optimization challenge at GECCO 2026, using permutation encodings and simulated annealing. Our solutions were published as two GECCO 2026 papers.

UPM Thesis LaTeX Template — 2024
A LaTeX thesis template for Universiti Putra Malaysia, built from the official Microsoft Word template.

International Timetabling Competition 2019 — 2022
An implementation of the International Timetabling Competition 2019 benchmark, the latest benchmark for university course timetabling, built as the core of my Ph.D.

Car Evaluation Dataset — 2021
A study of feature engineering and of how different classification methods perform on the Car Evaluation dataset.

Java Framework for Genetic Algorithms — 2019
A Java framework to implement evolutionary algorithms, apply them to different problems, and compare their performance. Built during my master's research.

Tic-Tac-Toe Game — 2018
A game with an AI opponent that uses Q-Learning and has several difficulty levels. Built for a Reinforcement Learning course.

Photo Noise Reduction — 2018
Reduces image noise with a message passing technique. Final project for a Probabilistic Graphical Models course.

Weather and Traffic Forecasting Software 2017
Forecasts weather and traffic density with a neural network trained on historical data. Built for an Artificial Neural Networks course.

Auto Parts Finder 2016
An Android app used by assembly line workers at IKCO to check parts stock in real time and order parts, which cut idle time on the line.

Skills

Operations Research: Combinatorial Optimization, Mixed Integer Programming, Stochastic Optimization, Constraint Programming, Metaheuristics, Linear and Mathematical Programming, Gurobi, Google OR-Tools

Data Science: Predictive Modeling, Supervised and Unsupervised Learning, Data Visualization, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Power BI, Tableau, Excel

Programming: Python, Java, SQL, MySQL, SQLite, XML, Git, GitHub, UML

Research and Teaching: Scientific Writing, LaTeX, Public Speaking, Presenting, Teaching

Languages: English (Bilingual, IELTS 8), Mandarin (Intermediate), Malay (Basic)

Certificates

Supervisory Training Certificate (SURE) 2026

Full Course Curriculum Design Workshop and Certificates for MOHE and CQM 2026
OBE, Table 4, ODL, and rubrics

Chinese (Mandarin) Certificate May 2021
Peking University

Google Data Analytics Certificate Apr 2021
Google

IELTS Academic, Overall Band 8 Jun 2019
IELTS

Certificate in Advanced English (CAE), C2 Oct 2016
Cambridge Assessment English

Cambridge Teacher Training Course (TTC) Jul 2016
Kish Institute of Science and Technology

Workshops and Seminars

Classifying Problems by Computational Complexity: P vs NP and Beyond, by Professor Paul Goldberg 2025

CORS Micro-event on Hybrid Quantum-Classical Computing for Internet of Everything Optimization, by Dr. Mahzabeen Emu 2025

Best Practices of Teaching Supply Chain Analytics to Meet Job Market Requirements 2025

Deutscher Akademischer Austauschdienst (DAAD) Webinar 2025

Google I/O 2025

CORS Transportation & Logistics SIG Seminar: The Future of Mobility, From Gridlock to Green Solutions, by Leandro Coelho 2025

CORS Micro-event on Public Transit and Supply Chains: How Data Can Support Optimization in Public Transit Operations, by Professor Martin Trepanier 2025

CORS Micro-event on Analytics: An OR Tale of Analytics, AI, and Auctions, by Professor Fredrik Ødegaard 2024

How to (Ethically) Increase Your Citations, by Professor Graham Kendall 2024

CORS SIG in Energy, Environment, and Sustainability: Optimization Challenges in Electric Vehicle Charging Operations, by Professor Miguel Anjos	2024
Expert Talk: Graph-based Explainable and Fair Recommendation, by Professor Guandong Xu	2024
2nd CORS Micro-Event on Circular Economy and Autonomous Mobility	2024
Tips and Guide for Publishing in High-Impact Journals	2024
Elsevier Workshop on Research Data Management: Dive Into the Details	2024
AI Tools for Literature Review Search and Writing	2024
Online Database Classes: Scopus	2024
United States Air Force Office of Scientific Research (AFOSR) International Grant Funding Briefing	2024
NVIDIA Research Workshop	2023
Postgraduate Colloquium, Universiti Putra Malaysia	2023
Research Showcase at FSKTM Open Day, Universiti Putra Malaysia	2023
Elsevier Young Researchers Workshop	2023
UPM Workshop on High-Impact Publishing	2022
IEEE Young Professionals Malaysia Seminar	2021
Workshop on How to Publish with Oxford Journals	2021

References

Professor Anwar P. P. Abdul Majeed
Sunway University, anwarm@sunway.edu.my

Professor Dr. Thomas Weise
Hefei University, tweise@hfuu.edu.cn

Associate Professor Dr. Razali Yaakob
Universiti Putra Malaysia, razaliy@upm.edu.my

Professor Salwani Abdullah
Universiti Kebangsaan Malaysia, salwani@ukm.edu.my

Professor Graham Kendall
University of Nottingham, graham.kendall@nottingham.ac.uk